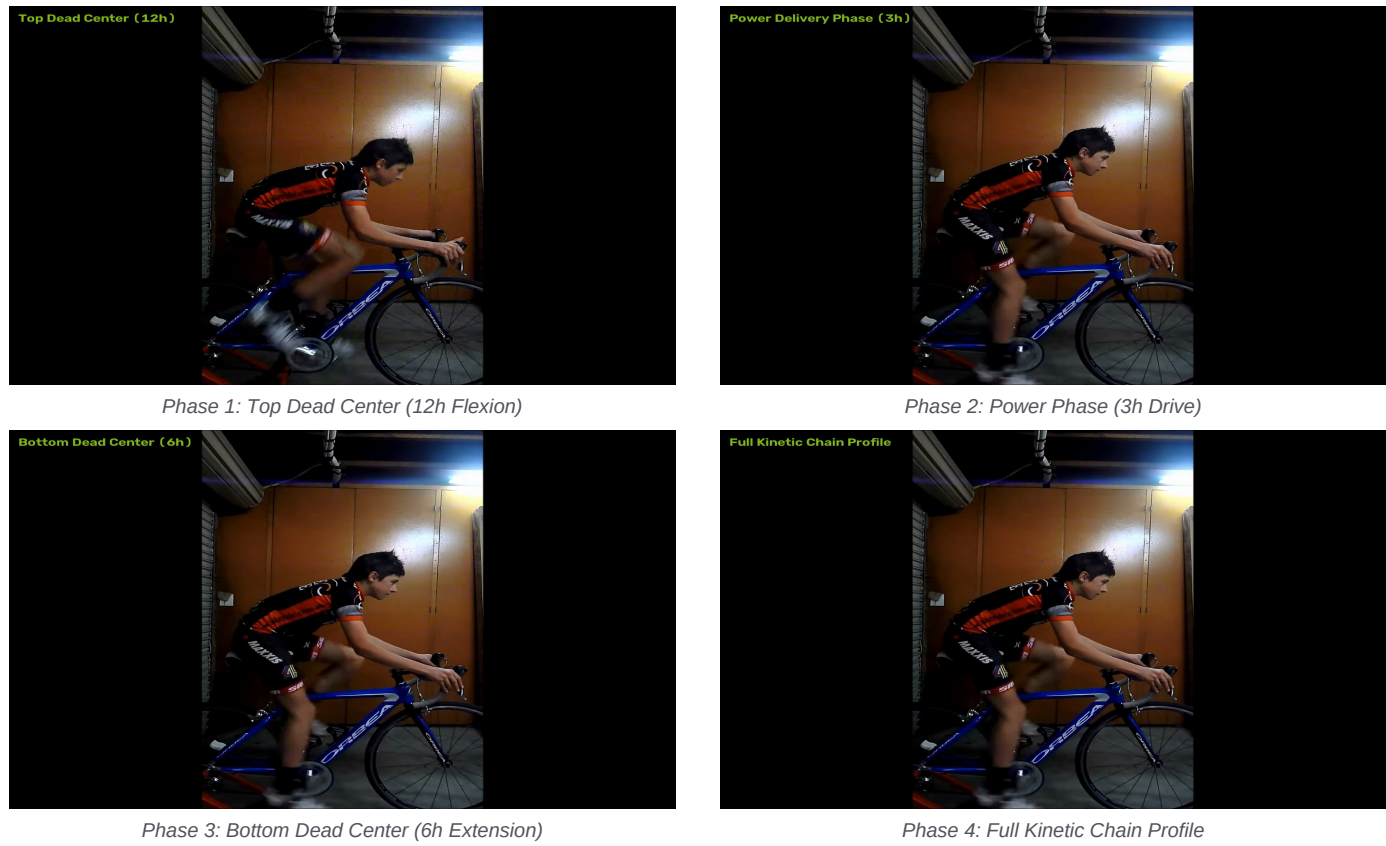


OPEN-BIKEFIT BIOMECHANICAL REPORT

Kinematic Posture Baseline & Hardware Adjustment Guide

Rider: Alex Chen	Discipline: ROAD	Target: Balanced Performance
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1. PEDAL STROKE DECOMPOSITION (4-PHASE KINEMATICS)



2. OBSERVED KINEMATIC ANGLES VS TARGET ENVELOPES

Kinematic Metric	Observed Value	Reference Target	Status
Knee Extension (BDC 6h)	146.2 deg	140 - 150 deg	Optimal
Knee Flexion (TDC 12h)	70.4 deg	68 - 75 deg	Optimal
Closed Hip Angle (TDC 12h)	48.8 deg	45 - 55 deg	Optimal
Torso Angle (to horizontal)	43.1 deg	40 - 50 deg	Optimal
Shoulder / Reach Angle	87.5 deg	85 - 95 deg	Optimal
Ankle Plantarflexion at BDC	96.2 deg	90 - 105 deg	Optimal

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3. STUDIO FIT SUMMARY & HARDWARE ADJUSTMENTS

Open-BikeFit Biomechanical Posture Report

Rider: `Alex Chen` | **Discipline:** `ROAD` | **Target Profile:** `Balanced Performance`
Overall Fit Alignment Score: `100% / 100%`

1. Kinematic Joint Angle Summary

- **Knee Extension at BDC (6 o'clock):** `146.2°` *(Target: 140°?150°)*
- **Knee Flexion at TDC (12 o'clock):** `70.4°` *(Target: 68°?75°)*
- **Closed Hip Angle at TDC:** `48.8°` *(Target: 45°?55°)*
- **Torso Angle (to horizontal):** `43.1°` *(Target: 40°?50°)*
- **Shoulder / Reach Angle:** `87.5°` *(Target: 85°?95°)*
- **Ankle Angle at BDC:** `96.2°` *(Target: 90°?105°)*

2. Prioritized Wrench Adjustments (Step-by-Step)

1. **Saddle Position:**
 - **Saddle Height:** Optimal (146.2° within 140°?150° window). Stable leg extension at BDC.
2. **Cockpit Stack & Reach:**
 - **Closed Hip Clearance:** Optimal (48.8° at 12 o'clock TDC).
 - **Cockpit Reach:** Balanced (87.5° arm-to-torso angle). Weight is neutrally supported.
3. **Pedal & Shoe Interface:**
 - **Ankling Mechanics:** Stable (96.2° at BDC).

3. Intake Symptom & Pain Point Diagnostics

- **Front of Knee (Patellar Discomfort):** Frequently caused by a saddle that is slightly too low or positioned too far forward, increasing knee flexion loads at the 3 o'clock power phase. Follow the saddle height adjustment above.
- **Lower Back Fatigue:** Common with excessive reach or tight closed hip angles. Increasing stem stack height by 5?10mm will reduce pelvis rotation torque.

4. Iteration & Adaptation Protocol

- Perform physical adjustments in single increments (e.g., 3?5 mm at a time).
- Test ride for 20?30 km under normal riding cadence before re-recording a progress video.
- *Notice: Open-BikeFit provides geometric kinematic baseline estimation to assist riders and bike fitters. It is not a medical diagnostic tool.*